Ponavljanje: Fourierov red (FS)

Vremenski kontinuirani Fourierov red

$$CTFS_{T_0}(f(t)) = F_k = \frac{1}{T_0}\int_{T_0} f(t)\cdot e^{-j\omega_0 k t}\,dt$$

koeficijent Fourierove transformacije

• postoje kad je definiran i konačan $\forall k \in \mathbb{Z}$

REKONSTRUKCIJA signala iz spektra F: ← ICTFS

$$f(t) = \sum_{k=-\infty}^{\infty} F_k \cdot e^{jk\frac{2\pi}{T_0}\cdot t}, \quad \forall t \in \mathbb{R}$$

→ $\{F_k\} \in \mathbb{Z}$ čine spektar signala (niz brojeva)

→ ako imamo niz brojeva (F_k) možemo natrag dobiti $f(t)$

→ f. period kontinuiranog signala: $f: \mathbb{R} \to \mathbb{R}\ (\mathbb{C})$

$$f(t+T_0) = \sum F_k \cdot e^{jk\frac{2\pi}{T_0}\cdot(t+T_0)} = \sum F_k \cdot e^{jk\frac{2\pi}{T_0}\cdot t}\cdot \underbrace{e^{jk2\pi}}_{1} = \sum F_k \cdot e^{jk\frac{2\pi}{T_0}} = f(t)$$

$\angle F$ = fazni spektar $\qquad |F|$ = amplitudni spektar

Signal konačne energije:

ako vrijedi $\int_{T_0} |f(t)|^2\,dt < \infty \Rightarrow \int_0^{T_0} \left|f(t) - \sum_{k=-K}^{K} F_k \cdot e^{jk\omega_0 t}\right|^2 dt \to 0$ za $K \to \infty$

Osnovna svojstva

za periodične signale konačne energije na periodi

$$P_f = \frac{1}{T_0}\int_{T_0} |f(t)|^2\,dt = \sum_{k=-\infty}^{\infty} |F_k|^2$$

izvod: $P_f = \frac{1}{T_0}\int_{T_0} f(t) * f(t)\,dt = \frac{1}{T_0}\int_{T_0} f(t)\sum_{k}^{\infty} F_k^* e^{-j\frac{2\pi}{T_0}kt}\,dt$

„minus" jer je konjugirano kompleks

$$= \sum_k F_k^* \frac{1}{T_0}\int_{T_0} f(t) e^{-j\cdot\frac{2\pi}{T_0}\cdot kt}\,dt = \sum_k F_k^* \cdot F_k = \sum_{k=-\infty}^{\infty} |F_k|^2$$

CTFS realnih signala

- spektar je konjugirano sim. $\rightarrow F_{-k} = F_k^*$
- amplitudni spektar je paran $|F_k| = |F_{-k}|$

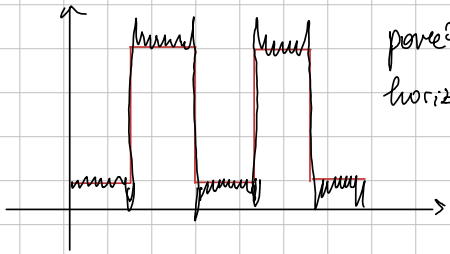
  fazni je neparan $\angle F_k = -\angle F_k$

$\Rightarrow$ kompaktni trig. oblik: $f(t) = F_0 + \sum_{k=1}^{\infty} 2|F_k| \cdot \cos(k\omega_0 t + \angle F_k)$

$\rightarrow$ najbolja aproksimacija u smislu najmanje energije pogreške:

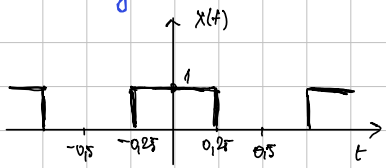
$$\hat{f}_K = \sum_{k=-K}^{K} F_k \cdot e^{jk\frac{2\pi}{T_0}t}$$

Gibbsova pojava:



povećanjem K bolje se poklapa horizontalno, ali dolazi do znatnih nadvišenja

Primjer: Zadano



$\rightarrow F_k = \dots = \frac{1}{2} \cdot \frac{\sin\left(\frac{\pi}{2}k\right)}{\frac{\pi k}{2}}$ *(sinc fja)

* $k=0$

$F_0 = \dots = \frac{1}{2}$

$\hookrightarrow$ budući da je zadana fja $(x(t)=1)$ parna $\rightarrow$ dobiveni F_k su $\in \mathbb{R}$

- Zadani signal je kontinuirana fja pa će njegov spektar biti aperiodičan
- Signal je periodičan, njegov spektar je diskretan